Jeonghyun Woo

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RESEARCH INTERESTS

My research spans **computer architecture**, **hardware/systems security**, **memory systems**, and **AI/ML systems**, with the goal of building computing systems that are both secure and efficient:

- **Secure and Efficient Memory Systems:** Building high-performance, energy-efficient, and secure memory systems across conventional and emerging technologies, with defenses against RowHammer, denial-of-service, and side-channel attacks.
- **Trustworthy AI Systems:** Designing lightweight cross-layer defenses against hardware-induced privacy and integrity threats in LLM serving, including bit-flip jailbreaks and side-channel leakage from MoE and speculative decoding.
- **Efficient LLM Inference Systems:** Co-designing algorithms and memory systems to improve performance and energy efficiency of LLM serving, including KV-cache management, speculative decoding, and energy-aware memory controllers.
- **Secure and Efficient Microarchitecture:** Improving the performance and energy efficiency of key microarchitectural components, such as prefetchers and branch predictors, across CPUs and GPUs, without introducing security vulnerabilities.

EDUCATION

The University of British Columbia (UBC) — September 2022 – November 2026 (Expected)
Ph.D., Electrical and Computer Engineering | GPA: 4.00/4.00 — *Vancouver, BC, Canada*
Advisor: Prof. Prashant Nair
Dissertation: *Secure and Low-Overhead Architectural Mitigations for Trustworthy and Scalable Memory Systems*

Hanyang University (HYU) — March 2018 – February 2020
M.S., Electronics and Computer Engineering | GPA: 4.00/4.00 — *Seoul, South Korea*
Advisor: Prof. Ki-Seok Chung
Thesis: *RowHammer Mitigation Architecture for Highly Reliable DRAM*

Hanyang University (HYU) — March 2012 – February 2018
B.S., Electronic Engineering, Summa Cum Laude | GPA: 3.89/4.00 — *Seoul, South Korea*

PROFESSIONAL EXPERIENCE

The University of British Columbia (UBC) — September 2022 – Present
Graduate Research Assistant, ***STAR Lab*** — *Vancouver, BC, Canada*
Advisor: Prof. Prashant Nair

NVIDIA Research — May 2024 – August 2024
Research Intern, ***Architecture Research Group*** — *Westford, MA, USA*
Mentor: Dr. Aamer Jaleel | Manager: Dr. David Nellans

Micron Technology — May 2023 – August 2023
Systems Research Engineering Intern, ***Vertical Systems Research*** — *Folsom, CA, USA*
Mentor: Dr. Chun-Yi Liu | Manager: Ameen Akel

Hanyang University (HYU) — March 2018 – February 2020
Graduate Research Assistant, ***ESoC Lab*** — *Seoul, South Korea*
Advisor: Prof. Ki-Seok Chung

HONORS AND AWARDS

Distinguished Artifact Award, HPCA 2025 — *1 of 2 Distinguished Artifacts*	2025
Best Paper Award, HPCA 2023 — *1 of 2 Best Papers from 364 submissions*	2023
Student Travel Grants — ISCA (2025, 2026); HPCA (2023, 2025)	2023, 2025, 2026
Faculty of Applied Science Graduate Award, UBC — $24,800 CAD total	2022, 2023, 2024, 2025
Hanyang Graduate School Scholarship — 70% tuition coverage (∼$9,200 CAD/yr)	2018, 2019
Outstanding Paper Award, KICS Fall Conference	2017
Hanyang Academic Excellence Award — Top 1% of ∼15,000 university students	2016, 2017
Hanyang Alumni Association Scholarship — Full tuition (∼$10,000 CAD/yr)	2016, 2017
Excellent Tutor Award, Engineering Mathematics Tutoring Program, Hanyang University	2016
Hanyang University Scholarship — Full tuition (∼$10,000 CAD/yr)	2012, 2013

PUBLICATIONS

Preprints and Under Review

[**P1**] "PrisonBreak: Jailbreaking Large Language Models with at Most Twenty-Five Targeted Bit-flips."
Zachary Coalson, **Jeonghyun Woo**, Chris S. Lin, Joyce Qu, Yu Sun, Shiyang Chen, Lishan Yang, Gururaj Saileshwar, Prashant J. Nair, Bo Fang, and Sanghyun Hong.
Submitted to *15th International Conference on Learning Representations* ***(ICLR'27)***, 2024.
arXiv

Peer-Reviewed Conference Publications

[**C6**] "Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense."
Jeonghyun Woo*, Junsu Kim*, Aamer Jaleel, and Prashant J. Nair. *(*equal contribution)*
53rd Annual International Symposium on Computer Architecture ***(ISCA'26)***, June 2026. *Acceptance Rate: 19.1%.*
Paper Code Slides

[**C5**] "When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations."
Jeonghyun Woo, Joyce Qu, Gururaj Saileshwar, and Prashant J. Nair.
52nd Annual International Symposium on Computer Architecture ***(ISCA'25)***, June 2025. *Acceptance Rate: 23.1%.*
Paper Code Slides
[Artifact Evaluated: Available, Functional, Reproduced]

[**C4**] "QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues."
Jeonghyun Woo, Shaopeng (Chris) Lin, Prashant J. Nair, Aamer Jaleel, and Gururaj Saileshwar.
31st International Symposium on High-Performance Computer Architecture ***(HPCA'25)***, March 2025. *Acceptance Rate: 21.0%.*
Paper Code Slides
[Distinguished Artifact Award — 1 of 2 Distinguished Artifacts]
[Artifact Evaluated: Available, Functional, Reproduced]

[**C3**] "DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense."
Jeonghyun Woo and Prashant J. Nair.
31st International Symposium on High-Performance Computer Architecture ***(HPCA'25)***, March 2025. *Acceptance Rate: 21.0%.*
Paper Slides

[**C2**] "Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems."
Jeonghyun Woo, Gururaj Saileshwar, and Prashant J. Nair.
29th International Symposium on High-Performance Computer Architecture ***(HPCA'23)***, February 2023. *Acceptance Rate: 25.0%.*
Paper Code Slides
[Best Paper Award — 1 of 2 Best Papers from 364 submissions]
[Artifact Evaluated: Available, Functional, Reproduced]

[**C1**] "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer."
Kwangrae Kim, **Jeonghyun Woo**, Junsu Kim, and Ki-Seok Chung.
39th International Conference on Computer Design ***(ICCD'21)***, October 2021. *Acceptance Rate: 24.4%.*
Paper Slides Video

Workshop Publications and Posters

[**W3**] "CnC-PRAC: Coalesce, Not Cache, Per-Row Activation Counts for an Efficient in-DRAM Rowhammer Mitigation."
Chris S. Lin, **Jeonghyun Woo**, Prashant J. Nair, and Gururaj Saileshwar.
Fifth Workshop on DRAM Security ***(DRAMSec'25)***, co-located with ISCA 2025, June 2025.
Paper

[**W2**] "Counterpoint: One-Hot Counting for PRAC-Based RowHammer Mitigation."
Shih-Lien Lu, **Jeonghyun Woo**, and Prashant J. Nair.
Fifth Workshop on DRAM Security ***(DRAMSec'25)***, co-located with ISCA 2025, June 2025.
Paper

[**W1**] "HammerFilter: Robust Protection and Low Hardware Overhead Method for Row-Hammering."
Kwangrae Kim, Junsu Kim, **Jeonghyun Woo**, and Ki-Seok Chung.
Work-in-Progress Poster, *58th Design Automation Conference* ***(DAC'21)***, December 2021.
Poster

Domestic (Korean) Conference Publications

[**D2**] "A Method to Find the Optimal Probability for Probability-Driven Additional Row Refresh to Prevent DRAM Row Hammering."
Jeonghyun Woo and Ki-Seok Chung.
Korean Institute of Communications and Information Sciences (KICS) Winter Conference, January 2019.

[**D1**] "Implementation of an FPGA-Based CNN Accelerator Using SDSoC."
Changwoo Lee*, **Jeonghyun Woo***, Sang-Soo Park, and Ki-Seok Chung. *(*equal contribution)*
KICS Fall Conference, November 2017.
Code **[★ 300+ GitHub stars]**
[Outstanding Paper Award]

INVITED TALKS

When Optimization Breaks Security: Architecting Efficient and Trustworthy Hardware
Nanyang Technological University, Singapore, July 2026

Built on Broken Ground: Why Secure Computing Must Start at the Hardware
University of Waterloo, Waterloo, Canada, March 2026

Towards Secure and Scalable Memory Systems: From RowHammer Attacks to Hardware-Induced LLM Jailbreaks
KAIST CASYS Seminar, Remote, November 2025

Towards Secure and Low-Overhead PRAC-Based RowHammer Mitigations
NVIDIA Research (end-of-internship talk), Westford, USA, August 2024

RowHammer Mitigations for Future High-Bandwidth Memory
Micron Technology (end-of-internship talk), Folsom, USA, August 2023

Conference Talks

Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense
ISCA'26, Raleigh, USA, June 2026

When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations
ISCA'25, Tokyo, Japan, June 2025

QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues
HPCA'25, Las Vegas, USA, March 2025

DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense
HPCA'25, Las Vegas, USA, March 2025

Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems
HPCA'23, Montreal, Canada, February 2023

Implementation of an FPGA-Based CNN Accelerator Using SDSoC
KICS'17 Fall, Daegu, South Korea, November 2017

PROFESSIONAL SERVICE

Program Committee

- 2026 IEEE/ACM International Symposium on Microarchitecture (**MICRO'26**) — Student PC
- 2026 IEEE International Symposium on High-Performance Computer Architecture (**HPCA'26**) — Light PC

Artifact Evaluation Committee

- 2025 IEEE International Symposium on Workload Characterization (**IISWC'25**)

Invited Reviewer

- IEEE Computer Architecture Letters (**CAL**), 2026
- IEEE Transactions on Very Large Scale Integration Systems (**TVLSI**), 2025

Student Volunteer

- 2025 SPICE Workshop, co-located with MICRO'25
- 2024 IEEE International Symposium on Workload Characterization (**IISWC'24**)
- 2023 International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'23**)

TEACHING EXPERIENCE

The University of British Columbia (UBC) Vancouver, BC, Canada

Graduate Teaching Assistant

CPEN 411: Computer Architecture 2022, 2023, 2024, 2026

CPEN 312: Digital Systems and Microcomputers 2025

Hanyang University (HYU) Seoul, South Korea

Graduate Teaching Assistant

ELE 3081: VLSI Engineering 2019

ITE 4003: SoC Design 2018

MENTORING EXPERIENCE

Junsu Kim January 2020 – Present

Ph.D. Student, The University of British Columbia (UBC) *Vancouver, BC, Canada*

Project: Scalable and robust RowHammer mitigations → **Published at:** DAC'21 (WIP), ICCD'21, ISCA'26

Project: Efficient LLM inference via KV-cache management and speculative decoding

Kwangrae Kim January 2019 – December 2021

Ph.D. Student, Hanyang University (HYU) *Seoul, South Korea*

Project: Low-cost and robust probabilistic RowHammer mitigation → **Published at:** DAC'21 (WIP), ICCD'21

SELECTED ACADEMIC PROJECTS

Compression-Assisted Last-Level Cache Replacement January 2023 – April 2023

Advanced Computer Architecture, UBC (Instructor: Prof. Mieszko Lis)

Showed that 72.7% of cache lines are compressible by $\geq$10 B; proposed a compression-assisted LLC replacement policy that improves hit rates with minimal metadata overhead.

Revisiting Address Translation on Intel Optane DC PMEM September 2022 – December 2022

Graduate Operating Systems, UBC (Instructor: Prof. Margo Seltzer)

Quantified address-translation overheads on Optane DC PMEM with graph processing, HPC, and genomics workloads; demonstrated that Transparent Huge Pages mitigate 4 KB page bottlenecks.

Modified LeNet-5 Forward Pass in CUDA November 2020 – December 2020

Applied Parallel Programming, UIUC

Implemented optimized convolutional forward passes using shared-memory tiling, constant memory, and loop unrolling; profiled with Nsight Systems and Nsight Compute.

TECHNICAL SKILLS

Programming: C/C++, Python, CUDA, Verilog, Perl, Bash, Go, Assembly

Simulators: Ramulator, ChampSim, gem5, DRAMSim2, GPGPU-Sim, MGPUSim

Hardware Tools: Pin, SimPoint, Xilinx Vivado, Xilinx SDSoC, Intel Quartus